

Ali Hassan

Embedded Systems Engineer | Firmware & Hardware Integration

📍 Lyon, France | ✉ alihassan.pk019@gmail.com | 📞 +33 7 45 57 81 66 | 💻 alihassan019 | 🌐 alihassan-019.github.io

Professional Summary

Embedded Systems Engineer with experience in firmware development, electronics integration, and real-time embedded applications. Experienced with STM32, nRF52, PIC microcontrollers, RTOS-based systems, communication interfaces, and sensor-driven platforms. Skilled in hardware-software integration through PCB design, schematic development, peripheral configuration, and system validation for robotics, IoT, and medical-device applications.

Technical Skills

Programming: Embedded C, C++, Python, MATLAB

Embedded Platforms: STM32 (ARM Cortex-M), nRF52, PIC, Jieli AC6966

Firmware & RTOS: FreeRTOS, Zephyr RTOS, Device Drivers, Interrupts, DMA, Firmware Development

Hardware: PCB Design, Schematic Design, Sensor Integration, ADC Interfaces, Hardware-Software Integration

Communication Protocols: UART, SPI, I2C, BLE, RS-485, Modbus RTU, MQTT, LoRaWAN, Zigbee

Robotics & Control: ROS 2, PID Control, Sensor Fusion, Localization, Motor Control, Encoder Feedback

Development Tools: Altium Designer, STM32CubeIDE, Keil uVision, MPLAB X, MATLAB/Simulink, Linux, Git, CI/CD

Professional Experience

Robotics & Embedded Systems Engineer Intern, ABMI Groupe

Lyon, France | Mar 2026 – Sep 2026

- Developed ROS 2 components for sensor integration, real-time communication, and autonomous coupling.
- Implemented control algorithms integrating UWB, IMU, and vision sensors for robotic alignment.
- Supported embedded integration, monitoring interfaces, and validation of a 6-DOF Stewart platform.

Embedded Systems Engineer, MindTune Innovations

Wah Cantt, Pakistan | Mar 2025 – Sep 2025

- Developed RTOS-based applications for connected devices, focusing on real-time operation and reliability.
- Designed wireless EEG acquisition platform integrating biosignal sensing, processing, and data transfer.
- Developed embedded audio firmware for wireless devices focusing on optimization and power efficiency.

Embedded Systems Engineer, Revive Medical Technologies

Islamabad, Pakistan | Aug 2023 – Mar 2025

- Developed embedded firmware for medical applications with control logic and real-time processing.
- Integrated sensors and wireless transmission for connected monitoring through hardware-software integration.

Projects

ISE to Modbus RTU Converter

🔗 GitHub

- Designed an industrial embedded converter integrating isolated power supply, ADC-based sensing, and RS-485 Modbus RTU communication for reliable sensor data acquisition.

Catheter Trackability Testing Machine

🔗 GitHub

- Developed an embedded medical testing system combining load-cell sensing, motor control, and firmware-based measurement processing for catheter evaluation.

EEG-Integrated Wireless Earbuds

🔗 GitHub

- Developed a wearable embedded platform integrating AC6966B Bluetooth audio firmware, ADS1299 biosignal acquisition, and nRF52 BLE communication.

Autonomous Coupling System Using a 6-DOF Stewart Platform

🔗 GitHub

- Built a ROS 2-based robotic platform integrating ArUco vision, IMU sensor fusion, motor control, and encoder feedback for autonomous alignment.

Education

Master's in Embedded Systems Security

Expected Sep 2026

Grenoble INP Esisar, Valence, France

Bachelor of Science in Electrical Engineering

Aug 2024

COMSATS University Islamabad, Pakistan

Languages

English: C1 | French: A1